Joanna quick start

Offline rowing IMU report - short setup and usage guide

This is the shortest workflow for creating the offline HTML dashboard from two recorded IMU CSV files. The output is a self-contained HTML file that can be opened in a browser and shared afterwards.

1. Clone the repository

```
git clone https://github.com/Miled-Mahdoui/AnIMU-Based-Wearable-Sensing-Approach-for-
Rowing.git
```

2. Open the project folder

```
cd AnIMU-Based-Wearable-Sensing-Approach-for-Rowing
```

3. Prepare the CSV files

Copy the SEAT and BOAT CSV files into one folder. The script can usually detect which file is SEAT and which file is BOAT from the first CSV column.

```
/path/to/folder_with_csv_files/
  SEAT_LOG.CSV
  BOAT_LOG.CSV
```

4. Create the HTML report

```
/usr/local/bin/python3 sketch_mar25a/analyze_log.py /path/to/folder_with_csv_files --output
rowing_report.html --label rowing_report --min-peak-ms 800 --smooth-window 10
```

5. Open the report

```
rowing_report.html
```

Recommended recording workflow

Start both IMUs before the athletes row. They do not have to start at exactly the same millisecond, but starting them close together makes the time notes easier. Let athletes row one after another, then write down approximate time windows from the beginning of the recording, for example Athlete 1: 00:10-01:15, Athlete 2: 01:30-02:40. Later, use the HTML dashboard to segment those time windows.

Short update about the analysis

The dashboard has changed: stroke detection now uses velocity-based drive-start detection instead of the older acceleration-peak based approach. In practice, a stroke is now detected from one drive-start movement to the next drive-start movement. This should make the segmentation more intuitive for coaches, because it follows the start of the rowing movement instead of focusing mainly on the largest acceleration spike.